



INTERNSHIP REPORT

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1 INTRODUCTION

This report documents my software engineering internship experience at Vihi IT Solutions, undertaken from 8th of September 2025 to 13th of March 2026. During this period, I was assigned to the Software Engineering department, where I gained extensive practical experience across a broad range of areas including frontend and backend development, database design, network infrastructure planning, system security, quality assurance, and technical documentation within an Agile environment.

My primary focus throughout the internship was the design, development, and delivery of the Mega Power Gym Management System, a comprehensive full-stack web application built to replace a fully manual gym management process. The system encompasses modules for member registration, staff management with role-based access control, attendance tracking, appointment booking, payment processing via Stripe, equipment maintenance, real-time chat, an AI-assisted chatbot, and a member portal with workout tracking and feedback features.

In addition to the Gym Management System, I contributed to several other significant projects: frontend development for the Vihi IT Solutions corporate website using React and Next.js; backend and frontend development for a team-based POS web application (PointSell) supporting both retail and restaurant operations; frontend development of the Marketplace and Boarding modules for the UniHub student platform; physical and logical network infrastructure design for the company's new three-storey office building using draw.io and Cisco Packet Tracer; and the authoring of a comprehensive user manual for the Delmege ERP system.

This report details my key activities, accomplishments, challenges encountered, and the valuable technical and professional skills acquired throughout this enriching internship experience.

1.1 Company Overview

Vihi IT Solutions is a growing software development and IT services company based in Sri Lanka. The company specialises in custom software development, mobile application development, IT consulting, and quality assurance testing. With a team of approximately 35 employees operating across development, administrative, and executive functions, the organisation follows a structured project-team model where developers are grouped by project. The company serves both local and international clients, delivering scalable and secure technology solutions tailored to their business needs. The internship was carried out within the Software Engineering department, where interns are embedded directly into active project teams. The department follows an agile development methodology, with daily standups, sprint-based planning, and regular code reviews forming the core of its workflow.

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1.2 Project Overview

During the internship period (September 2025 to March 2026), four major projects were undertaken at Vihi IT Solutions:

1. Vihi IT Solutions Company Website (React / Next.js / Material-UI): Contributed as a frontend developer to the company's corporate website. Responsibilities included designing and

implementing multiple pages (Technology, Marketing, Design), ensuring responsive layouts across all breakpoints, and maintaining visual consistency in line with the company's brand guidelines. The team consisted of a senior frontend developer (Mr. Dumindu) and the intern.

2. Mega Power Gym Management System (React / Node.js / Express / MySQL / Prisma ORM): Solely responsible for the full-stack development of a comprehensive gym management system. The system includes modules for member registration, staff management, attendance tracking, payment processing (Stripe integration), appointment booking, equipment management, real-time chat, chatbot support, and an announcement system. The intern served as the primary developer for both frontend and backend, with oversight from supervisor Mr. K.M.V.T. Bandara.
3. POS Web Application – PointSell (React / Node.js / Express / MySQL): Joined a team project as a backend and frontend developer. Responsibilities covered database design, JWT authentication, RBAC middleware, inventory management APIs, sales transaction processing, restaurant table management, kitchen display system, payment gateway integration (Stripe and PayHere), offline-first sync, and comprehensive testing using Jest and Cypress. The team was led by H.M.L.P. Abayarathne.
4. UniHub Platform – Marketplace and Boarding Modules (Next.js / TypeScript): Worked on the Marketplace module (product browsing, cart, checkout, vendor onboarding, order management) and the Boarding module (listing display, search/filter, detail pages) within an existing multi-module student platform. Collaborated with Mr. Gayan Pathirage as the lead developer.

This diverse project portfolio provided me with a well-rounded internship experience, allowing me to develop a broad range of technical skills and contribute meaningfully to Vihi IT Solutions. The following sections will provide a more in-depth look at each of these projects.

1.3 Glossary

Following are the abbreviations and definitions used in this document. Include abbreviations that are specific to the report's purpose

Abbreviation	Description
IT	Information Technology
ORM	Object-Relational Mapping
API	Application Programming Interface
JWT	JSON Web Token
RBAC	Role-Based Access Control
MVC	Model-View-Controller
POS	Point of Sale
UI/UX	User Interface / User Experience
SRS	System Requirements Specification
QA	Quality Assurance
KDS	Kitchen Display System
PWA	Progressive Web Application
CI/CD	Continuous Integration / Continuous Deployment
SLIIT	Sri Lanka Institute of Information Technology
ERP	Enterprise Resource Planning

2 INTERNSHIP INSIGHT

2.1 Objectives

2.1 Objectives This internship at the Bank of Ceylon's Data Centre was designed with a set of mutual objectives, benefiting both the company and myself.

Company Objectives:

- **Project Completion:** The Bank of Ceylon aimed to leverage my skills and knowledge to contribute to the development and deployment of the Data Center Inventory Management System (DCIMS), a critical project for modernizing their tape management processes. They also sought assistance with various system administration tasks to maintain smooth data center operations.
- **Fresh Perspective:** The company sought a fresh perspective on existing processes, hoping to identify areas for improvement and innovation through the intern's contributions.
- **Skill Development and Evaluation:** The internship program provided the company with an opportunity to evaluate potential future employees and identify talented individuals for their workforce.

My Objectives:

- **Practical Experience:** My primary objective was to gain hands-on, practical experience in software development within a real-world, professional setting. I aimed to apply theoretical knowledge gained in my academic studies to solve real-world problems and develop practical skills.
- **Skill Enhancement:** I sought to enhance my skills in specific technologies relevant to the projects, including React, Node.js, Python, SQL, and system administration. I also aimed to improve my abilities in database management and agile development methodologies.
- **Professional Development:** Beyond technical skills, I aimed to develop essential professional skills, including communication, teamwork, problem-solving, and time management within a corporate environment.
- **Career Exploration:** The internship provided a valuable opportunity to explore a career in data center management and software engineering within the financial sector, allowing me to gain a deeper understanding of the industry and its specific challenges. These shared objectives formed the basis of my internship experience, guiding my work and providing a framework for evaluating the internship's overall success. The degree to which these objectives were met will be addressed in subsequent sections of this report.

2.2 Procedures

My internship at Vihi IT Solutions followed a well-defined structure, ensuring clear communication, efficient task management, and continuous feedback.

Daily Workflows:

- **Task Assignment and Tracking:** Tasks were primarily assigned and tracked using internal project management tools (Jira). This allowed for clear task definitions, assignment of responsibilities, progress tracking, and documentation of any issues or roadblock encountered. Mr. K.M.V.T. Bandara, my supervisor, prioritized tasks based on project needs and sprint goals. For smaller, ad-hoc tasks (like documentation updates or quick bug fixes), communication was usually done directly or via messaging.
- **Daily Reporting:** I maintained a daily diary (Form I-3A) detailing the activities performed, challenges faced, and progress made on each project. This log served as a record of my daily contributions and provided a basis for discussions during team meetings.
- **Agile Scrum Framework (for PointSell POS System):** The development of the PointSell POS system followed the Agile Scrum framework. I participated in daily stand-up meetings where the team discussed progress, identified roadblocks, and planned for the day's activities. We also conducted regular sprint reviews to demonstrate completed features and gather feedback from stakeholders.
- **Frontend and Backend Development Tasks:** Development tasks for the Gym Management System, company website, and UniHub platform were managed through feature-based Git branching. Each task was broken down into subtasks before implementation, and code was submitted through pull requests for review before merging.

Performance Evaluation:

- **Regular Feedback:** I received regular informal feedback from my supervisor on my performance and progress. This feedback was invaluable in helping me improve my skills and address areas for development.
- **Progress Reviews:** Weekly progress review meetings were held with the supervisor to evaluate the quality of work, identify areas for improvement, and plan upcoming tasks. At key milestones, formal demonstrations of completed modules were presented to Mr. Bandara and senior team members (e.g., Mr. Dumindu and Ms. H.M.L.P. Abayarathne). Performance was continuously evaluated based on code quality, responsiveness to feedback, initiative, and adherence to professional standards.

- **Final Evaluation:** A final evaluation was conducted at the end of the internship, which assessed my overall performance and contributions to the projects. This evaluation was based on factors like project completion, technical skills demonstrated, problem-solving abilities, teamwork, communication, and overall adherence to the internship objectives.

These procedures and workflows ensured that my internship was productive, well-organized, and provided consistent opportunities for learning and improvement. The structure promoted effective communication with my supervisor and the development team, enabling me to contribute effectively to the projects while gaining valuable experience in a professional environment.

2.3 Methodology

Throughout my internship, I employed a variety of methods, tools, and techniques to effectively carry out my tasks and contribute to project success.

Software Development (Mega Power Gym Management System):

- **Full-Stack Development Approach:** The Gym Management System was developed as a complete full-stack application using an iterative development approach. I gathered requirements from the gym owner, designed the database schema, implemented backend APIs, and developed the frontend user interface. Regular feedback sessions were conducted with Mr. K.M.V.T. Bandara to ensure alignment with project goals.
- **Version Control (Git):** Git, using GitHub, was essential for managing the Gym Management System source code. It allowed for version tracking, feature branching, merging, and efficient code management throughout the development lifecycle.
- **Project Management (Jira):** Jira was used for task management, bug tracking, and progress monitoring. It provided a centralized platform for organizing project activities, assigning tasks, tracking progress, and managing deadlines.
- **Integrated Development Environment (IDE):** Visual Studio Code was my primary development environment for writing, debugging, and testing code, with extensions for React development, Prettier for code formatting, and Git integration.
- **Database Design:** Used draw.io for ER diagramming to visualize database structures before implementation. Prisma ORM was used for type-safe database access and migration management.

PointSell POS System Development:

- **Agile Scrum:** The PointSell POS project followed the Agile Scrum methodology, enabling iterative development, continuous feedback, and adaptability to changing requirements. This involved:
 - **Sprint Planning:** Defining sprint goals, prioritizing tasks, and estimating effort.
 - **Daily Stand-up Meetings:** Tracking daily progress, identifying roadblocks, and promoting team communication.
 - **Sprint Reviews:** Demonstrating completed features to stakeholders and gathering feedback.
 - **Sprint Retrospectives:** Reflecting on the past sprint, identifying areas for improvement, and refining the development process.
- **Version Control (Git):** Git, using GitHub, was used for collaborative development with the team. Feature-based branching was used throughout, with pull requests reviewed before merging to the main branch.
- **Project Management (Jira):** Jira was used for task management, bug tracking, and sprint planning. It provided a centralized platform for organizing project activities, assigning tasks, tracking progress, and managing deadlines.
- **Testing Tools:** Jest was used for unit and integration testing of backend APIs. Cypress was configured for end-to-end (E2E) testing of critical user workflows. Postman was used throughout for manual API testing.
- **API Documentation:** Swagger/OpenAPI was used to generate interactive API documentation for the backend services.

Vihi IT Solutions Company Website Development:

- **Frontend Development:** Used React and Next.js for building responsive web pages. Material-UI component library was used for consistent UI styling. Figma was used for wireframing and UI design planning prior to implementation.
- **Responsive Design:** Applied mobile-first approach with CSS Grid, Flexbox, and media queries to ensure seamless functionality across various viewport sizes and devices.
- **Code Review Process:** Participated in code reviews with senior front-end developer Mr. Dumindu, receiving feedback on coding standards, component structure, and best practices.

UniHub Platform Contribution:

- **Marketplace Module Development:** Developed product browsing, cart functionality, checkout process, vendor onboarding, product management, and order management features using Next.js and TypeScript.
- **Boarding Module Development:** Implemented listing display, search/filter functionality, detail pages, and API integration. Used Figma for wireframing to plan layout and user flow before implementation.
- **Quality Assurance:** Conducted comprehensive manual testing across both Marketplace and Tutor workflows, identifying and documenting UI inconsistencies and validation errors for the development team.

Network Infrastructure Design:

- **Physical Network Design:** Used draw.io to create detailed physical floor plans for the company's new three-story office building (18m × 12m footprint across three floors). Designed network layouts including data outlets, access points, equipment racks, servers, printers, and displays.
- **Logical Network Simulation:** Used Cisco Packet Tracer to simulate network topology and validate connectivity requirements.
- **Cost Estimation:** Prepared Bill of Materials (BOM) covering active components (switches, routers, access points, firewall, servers), passive components (Cat6 UTP cabling, patch panels, cable trays), and end devices. Verified all cable runs were within the 802.3 maximum of 90m.

Documentation and Technical Writing:

- **User Manual Development:** Authored comprehensive user manual for the Delmege ERP system, covering 10 core modules including Dashboard, RAQ Management, Sectors Management, Policy Management, Client Company Management, Insurance Company Management, User Management, NatCat Exposure Dashboard, Settings, and Job Management.
- **Technical Documentation:** Created SETUP_GUIDE.md and DATABASE_SETUP.md for the Gym Management System, documenting environment setup, database schema design decisions, and migration processes.

These methods, tools, and techniques enabled me to work efficiently, collaborate effectively with the team, and contribute meaningfully to the projects during my internship. They also provided valuable experience with industry-standard practices and software development workflows.

3 LEARNING OUTCOME

3.1 Learning

My internship at Vihi IT Solutions provided extensive learning across both technical and non-technical domains.

Technical Learning:

- **Full-Stack Development:** Gained practical experience with the complete software development stack, from database schema design and backend API development to responsive frontend implementation and deployment preparation. This included working with React, Next.js, Node.js, Express.js, MySQL, Prisma ORM, and Sequelize across multiple real-world projects. I deepened my understanding of front-end and backend development principles, API design, database interactions, and the importance of a well-structured application architecture.
- **Database Design and Management:** Honed my database design skills, learning to create efficient database schemas, write complex SQL queries, optimize database performance using indexes, and manage schema migrations. I gained practical experience working with MySQL, Prisma ORM for the Gym Management System, and Sequelize ORM for the POS system.
- **Security Engineering:** Developed a strong foundation in application security, implementing JWT-based stateless authentication with refresh token mechanisms, bcrypt password hashing with a cost factor of 10, Role-Based Access Control (RBAC) middleware to restrict endpoint access based on user roles, OTP verification for secure transactions, Stripe webhook signature verification for payment confirmation, input sanitisation against SQL injection and XSS attacks, rate limiting to prevent brute force attacks, and audit logging using Winston and Sentry for real-time error tracking.
- **Payment Integration:** Integrated Stripe payment gateway for card payments with 3D Secure support and webhook handling for payment confirmation and membership activation. Integrated PayHere for local Sri Lankan mobile payment options in the POS system.
- **Networking and Infrastructure:** Designed physical network floor plans using draw.io for a three-storey office building (18m × 12m footprint), identifying key functional areas and network requirements. Implemented logical topologies with Cisco Packet Tracer for network simulation, including cost estimation and Bill of Materials (BOM) preparation covering active components (switches, routers, access points, firewall, servers), passive components (Cat6 UTP cabling, patch panels, cable trays), and end devices.

- **Testing and Quality Assurance:** Acquired hands-on experience with integration testing using Jest for backend API testing, end-to-end (E2E) testing using Cypress for critical user workflows, manual QA testing across browsers and devices, and CI/CD pipeline configuration for automated test runs.
- **Technical Documentation:** Learned how to write professional technical documentation including API guides using Swagger/OpenAPI for interactive API documentation, setup guides (SETUP_GUIDE.md, DATABASE_SETUP.md) for project onboarding, database schema documentation with ER diagrams, and comprehensive user manuals (notably the Delmege ERP User Manual covering 10 core modules).
- **Frontend Development:** Developed proficiency in React (hooks, Context API, useReducer for complex state management), Next.js (server-side rendering), Material-UI component library, responsive design principles (mobile-first approach, CSS Grid, Flexbox), and UI prototyping using Figma.

Non-Technical Learning:

- **Professional Communication:** Participated in client meetings with gym owners for requirement gathering, daily standups with the development team, sprint reviews for feature demonstrations, and supervisor progress evaluations with Mr. K.M.V.T. Bandara. This developed my ability to clearly present technical decisions, articulate implementation approaches, and receive constructive feedback professionally.
- **Team Collaboration:** Worked within cross-functional teams, co-developing modules with colleagues (e.g., POS system with Ms. H.M.L.P. Abayarathne) and integrating work across frontend and backend teams. Participated in code reviews, pair programming sessions, and collaborative problem-solving.
- **Problem-Solving and Adaptability:** Demonstrated versatility by transitioning between distinctly different project types (web development for Gym Management System, infrastructure design for office building network, system documentation for Delmege ERP) and quickly adapting to new tools and domains. Developed systematic approaches to debugging and troubleshooting.
- **Time Management and Task Prioritisation:** Managed multiple concurrent responsibilities across different projects, breaking down complex features into manageable subtasks, delivering iteratively within sprint cycles, and balancing academic leaves and holidays while maintaining project momentum.
- **Professionalism and Work Ethic:** Gained insights into professional conduct, accountability, and responsibility in a corporate setting. Understood the importance of meeting deadlines, maintaining code quality, and contributing to team success.

This comprehensive learning experience has equipped me with a diverse skill set and valuable experience that will be instrumental in my future career. The combination of technical expertise and professional development gained during this internship has prepared me to contribute meaningfully to the field of software engineering.

3.2 Measurable Outcome

This internship yielded several quantifiable outputs and demonstrable skill gains, providing concrete evidence of its impact on my professional development.

Artefacts Produced:

- **Vihi IT Solutions Website:** A fully functional corporate website deployed and ready for use. This system encompasses 5+ fully responsive production-ready web pages (Technology, Marketing, Design, Portfolio, Careers) built using React, Next.js, and Material-UI, automating the company's online presence and improving brand visibility.
- **Mega Power Gym Management System:** A complete full-stack web application developed to automate gym operations. This system encompasses features for member registration, staff management with role-based access control (RBAC), attendance tracking, appointment booking, equipment management, payment processing (Stripe integration), real-time chat, chatbot support, announcement broadcasting, member portal with workout tracking, and rating/feedback system. The system comprises 12+ database tables, 30+ React components, and a Node.js/Express backend with Prisma ORM, automating multiple manual processes and improving operational efficiency.
- **PointSell POS System:** A comprehensive Point of Sale web application supporting both retail and restaurant operations. This system features 45+ RESTful API endpoints across authentication, inventory, sales, payments (Stripe, PayHere), restaurant table management, kitchen order routing, offline sync, and reporting modules. The frontend includes a cashier dashboard, Kitchen Display System (KDS), inventory management UI, and analytics dashboard with Chart.js, streamlining business operations for end-users.
- **UniHub Platform Contributions:** Marketplace module (product browsing, cart, checkout, vendor onboarding, order management) and Boarding module (listing display, search/filter, detail view) implemented in Next.js/TypeScript, enhancing the platform's functionality for students and vendors.
- **Delmege ERP User Manual:** A professionally formatted, comprehensive user manual covering 10 core modules of the Delmege ERP/Broker Management System, including Dashboard, RAQ Management, Sectors Management, Policy Management, Client Company Management,

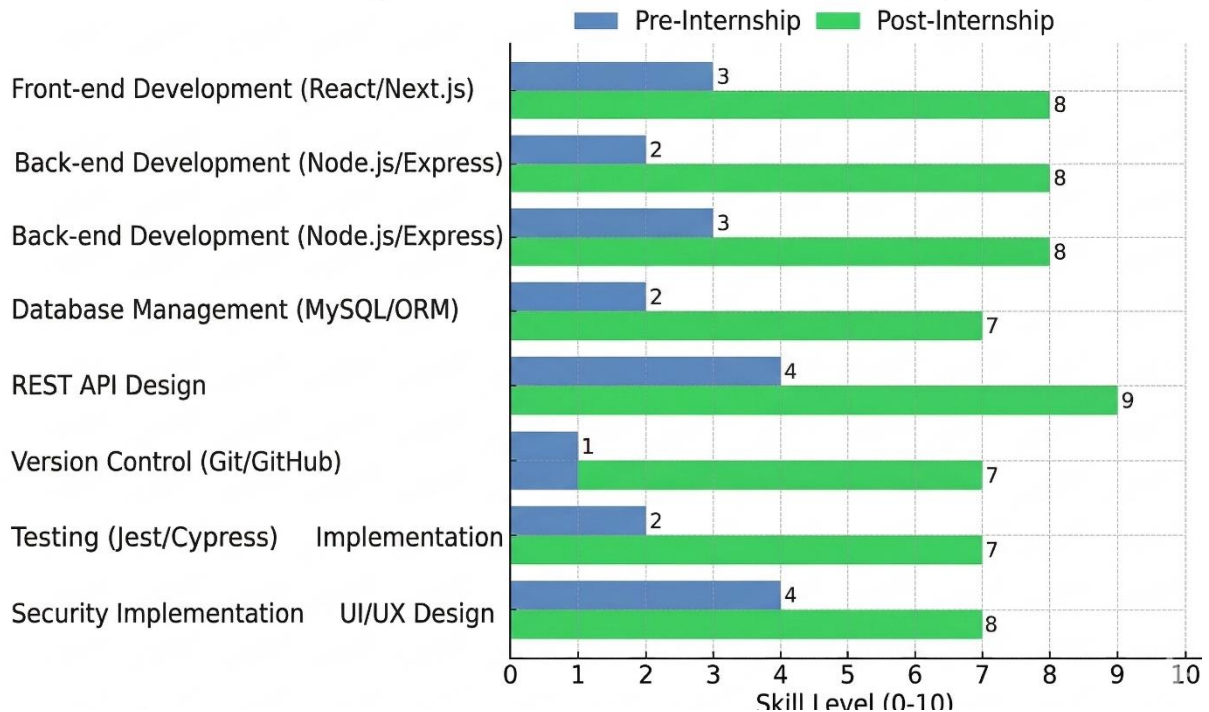
Insurance Company Management, User Management, NatCat Exposure Dashboard, Settings, and Job Management, facilitating user adoption and system understanding.

- Network Infrastructure Design: Physical floor plans (3 floors, 18m × 12m) and logical topology diagram for Vihi IT Solutions' new office building, including Cisco Packet Tracer simulation and Bill of Materials (BOM) covering active components, passive components, and end devices, ensuring proper network infrastructure planning.
- Technical Documentation: SETUP_GUIDE.md and DATABASE_SETUP.md for the Gym Management System, and Swagger/OpenAPI documentation for the POS system, ensuring project maintainability and developer onboarding.

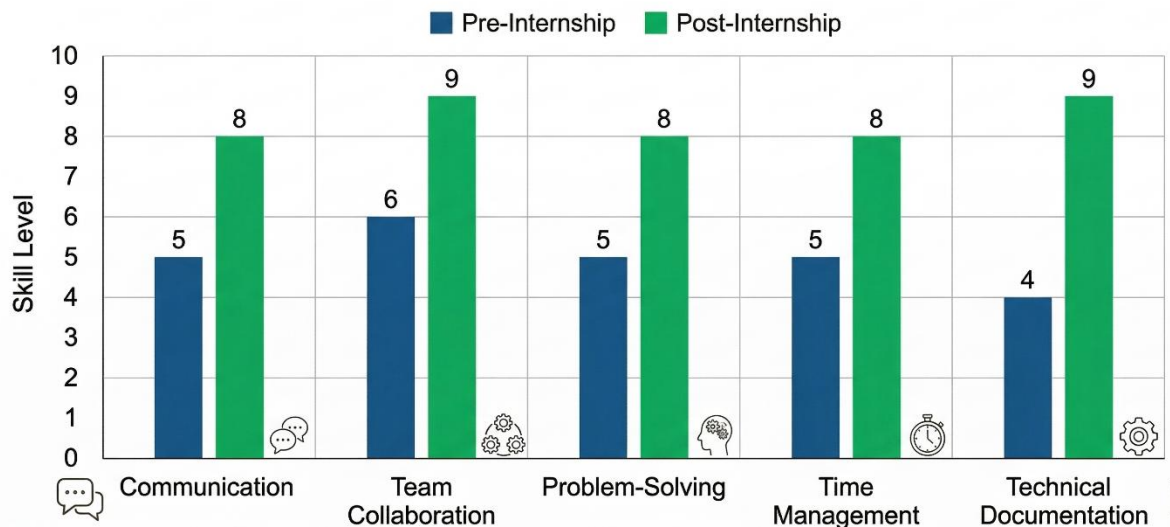
Comparative Skill Gains:

To visualize the progress made during my internship, the following bar chart and table provide a comparative view of the technical skills gained during the internship. Skills such as front-end development, back-end development, database management, API design, testing, security implementation, and version control were assessed based on a proficiency scale from 0 to 10, highlighting the improvement in each area.

Technical Skills Progression: Pre- vs. Post-Internship (Scale: 0-10)



Soft Skills Progression and Specific Skill Improvements (Scale: 0-10)



Skill Progression Timeline:

- Month 1 (September): Development activities were limited to orientation, environment setup, and reading code. Successfully completed onboarding, set up local development environment, and familiarized with Next.js and React technology stack.
- Month 2 (October): Full features were being independently designed and implemented. Successfully completed requirement gathering, database design, and backend setup for the Gym Management System.
- Month 3-4 (November-December): Complex security features, payment integrations (Stripe, PayHere), and real-time systems (chat, chatbot) were delivered. Completed the Gym Management System and contributed significantly to the POS system.
- Month 5-6 (January-March): Production-quality frontend systems with comprehensive testing (Jest, Cypress) and documentation (Swagger/OpenAPI, user manuals) were produced. Completed POS system, contributed to UniHub platform, and delivered network infrastructure design and ERP user manual.

Supervisor Evaluations Progression:

Supervisor evaluations consistently progressed from "highly positive and proactive start" and "excellent foundational skills" (Week 1) to "outstanding intern" and "highly valuable contributions" (final week), evidencing clear and measurable growth over the internship period.

Period	Supervisor Comment Summary
Week 1	Highly positive and proactive start; successfully completed onboarding and environment setup
Week 2	Highly productive; successfully applied Material-UI components for responsive design
Week 3	Excellent start to Gym Management project; strong analytical and communication skills
Week 4	Strong backend capabilities; effectively set up Node.js/Express with MVC pattern and Prisma ORM
Week 5	Rapid and highly effective frontend progress; successfully delivered core React components
Week 6	Delivered highly impactful features including chatbot and announcement broadcasting system
Week 7	Successfully concluded Gym project; impressive UI overhaul and dedication to code refactoring
Week 8	Strong adaptability and communication; effectively gathered requirements for POS system
Week 9	Strong technical capabilities in database and API design; implemented secure JWT authentication
Week 10	Significant strides in feature implementation; integrated diverse payment options
Week 11	Excellent technical initiative; implemented offline-first support with IndexedDB
Week 12	Proactively enhanced security; implemented rate limiting, Helmet.js, and input sanitization
Week 13	Excellent frontend capabilities; developed Kitchen Display System and inventory management
Week 14	Focused on system reliability; successfully built settings module and setup wizard

Period	Supervisor Comment Summary
Week 15	Successfully finalized POS tasks; implemented structured error handling and comprehensive documentation
Week 16	Strong analytical skills; thoroughly reviewed E-Learning architecture and identified issues
Week 17	Excellent full-stack capabilities; finalized vendor-side order management and cross-module testing
Week 18	Strong system design skills; implemented boarding feature with intuitive filtering options
Final Week	Outstanding intern; conducted thorough end-to-end testing; contributions highly valuable

Company's Inclusive Approach to Engineering:

Vihi IT Solutions maintains an open and collaborative working environment that actively promotes equality, diversity, and inclusion within its engineering practices. I observed this commitment through:

- **Diverse Teams:** The development teams reflected diversity in terms of gender, educational backgrounds, and professional experiences. Team members from diverse backgrounds contributed equally to projects, bringing a wide range of perspectives and innovative problem-solving approaches.
- **Equal Opportunities:** The company ensured equal access to mentorship, task assignments, and learning opportunities regardless of gender, background, or experience level. Female interns were given the same complexity of tasks and leadership responsibilities as their peers, reflecting the company's commitment to gender equality in engineering.
- **Respectful Communication:** A culture of respectful communication and collaboration was fostered within the team, where all voices were heard and valued. Junior members, including interns, were invited to contribute ideas and technical decisions on equal footing with senior developers.
- **Mentorship and Support:** Mentorship and support from senior team members (Mr. Dumindu, Ms. H.M.L.P. Abayarathne, Mr. Gayan Pathirage) were readily available to all interns, fostering a positive and inclusive learning environment.

- **Accessibility Considerations:** In UI development, the company encouraged implementation of accessibility features including keyboard navigation, screen reader support, and clear focus indicators, ensuring software was usable by people with diverse abilities.
- **Flexible Work Arrangements:** The company accommodated academic commitments, allowing interns to take leave for mid-term examinations and final exams while ensuring project continuity, demonstrating respect for diverse life circumstances.

Vihi IT Solutions' initiatives contribute to a more creative, innovative, and productive work environment. The company recognizes that diverse teams bring a wider range of perspectives and problem-solving skills, ultimately leading to better outcomes for the business and its customers. By actively promoting equality and inclusion, the company ensures that all team members feel valued, respected, and empowered to contribute their best work.

3.3 Effectiveness of the Effort and Reflection

This internship at Vihi IT Solutions proved highly effective in bridging the gap between academic learning and real-world application. The experience significantly enhanced my understanding of key concepts and provided practical context to the classroom material I have studied.

Connecting Classroom Knowledge to Practical Experience:

- **Database Design:** My coursework in database management systems provided a solid foundation in database design principles, including entity-relationship modeling, normalization forms, and SQL query writing. The internship allowed me to apply this knowledge practically, designing and implementing the database schemas for the Gym Management System and PointSell POS System. I learned about real-world considerations such as data integrity constraints, transaction management for concurrent operations, performance optimization through strategic indexing, and handling of edge cases such as race conditions during payment processing—concepts that were reinforced through hands-on implementation with Prisma ORM and Sequelize ORM.
- **Software Engineering Principles:** The internship reinforced the importance of software engineering principles like modularity, code reusability, and maintainability. Working on large-scale projects like the Gym Management System and POS system highlighted the practical benefits of these principles in creating robust and scalable applications. The MVC (Model-View-Controller) architectural pattern implemented in the POS backend solidified my understanding of separation of concerns. Code review practices taught me the importance of writing clean, readable, and well-documented code that can be maintained by other developers.
- **Agile Development:** The practical experience of working in an Agile Scrum team during the PointSell POS project solidified my understanding of agile principles learned in the classroom. I experienced firsthand the benefits of iterative development, continuous feedback, and

collaborative teamwork. Participating in daily stand-ups, sprint planning sessions, sprint reviews, and retrospectives gave me practical insight into how agile methodologies improve project transparency, adaptability, and team alignment.

- **Web Development:** While coursework introduced HTML, CSS, and JavaScript fundamentals, my internship exposed me to modern frameworks and libraries including React, Next.js, Node.js, and Express.js. I learned about responsive design implementation, state management patterns (useState, useReducer, Context API), server-side rendering with Next.js, RESTful API design, and frontend-backend integration—skills that are industry expectations but rarely covered comprehensively in academic settings.
- **Computer Networks:** My networking coursework provided theoretical understanding of OSI model, IP addressing, subnetting, and network topologies. The physical network floor plan design task for Vihi IT Solutions' new office building allowed me to apply this knowledge practically, considering cable length constraints (802.3 90m maximum for Cat6 UTP), equipment placement for optimal coverage, departmental connectivity requirements, and network security considerations. The Cisco Packet Tracer simulation further reinforced logical network design concepts.
- **Security Engineering:** Academic modules on cybersecurity introduced concepts like authentication, authorization, and encryption. The internship allowed me to implement these concepts practically through JWT-based stateless authentication, bcrypt password hashing with appropriate cost factors, Role-Based Access Control (RBAC) middleware, OTP verification for secure transactions, input sanitization to prevent SQL injection and XSS attacks, and rate limiting to prevent brute force attacks. These implementations gave me a deep understanding of security challenges in real-world applications.
- **Software Testing:** While classroom learning covered theoretical testing concepts, the internship provided hands-on experience with Jest for unit and integration testing, Cypress for end-to-end testing, and Postman for manual API testing. I learned to write test cases that cover edge scenarios, set up test databases with seed data, and configure CI/CD pipelines for automated test runs.

Improvements for Future Internships:

If presented with the same internship opportunity again, I would focus on the following improvements:

- **Deeper Dive into Specific Technologies:** While I gained experience across a wide range of technologies, I would dedicate more time to deepening my expertise in specific areas. For example, I would focus on advanced React concepts such as custom hooks, performance optimization techniques (useMemo, useCallback, React.memo), and advanced state management patterns. Alternatively, I would explore advanced database optimization techniques like query optimization, partitioning, and replication strategies.

- **Proactive Exploration of System Architecture:** I would take more initiative to explore the overall system architecture beyond my immediate tasks, understanding how different components interact, the reasoning behind architectural decisions, and the deployment infrastructure. This could involve shadowing senior developers during deployment processes, researching cloud infrastructure options, or exploring advanced monitoring and logging tools.
- **Enhanced Documentation Practices:** Although I created technical documentation including `SETUP_GUIDE.md`, `DATABASE_SETUP.md`, and Swagger/OpenAPI documentation, I would further improve my documentation practices by creating more comprehensive API specifications with detailed request/response examples, system architecture diagrams, and contribution guides for future developers. I would also consider creating video tutorials for complex features to aid user onboarding.
- **Early Stakeholder Engagement:** While we gathered feedback throughout projects, I would prioritize engaging with stakeholders even earlier in the design phase to ensure closer alignment with their needs and expectations from the outset. This would involve creating wireframes or prototypes for validation before implementation, scheduling regular check-ins during development, and establishing clear communication channels for feedback.
- **Automated Testing from Project Initiation:** I would set up automated testing frameworks (Jest, Cypress) at the very beginning of each project rather than adding them later. This would ensure better test coverage from the outset, reduce technical debt, and make it easier to maintain test suites as features are added.
- **Knowledge Sharing and Collaboration:** I would be more proactive in sharing learnings with team members through internal presentations or knowledge-sharing sessions. This would not only reinforce my own understanding but also contribute to the team's collective knowledge and foster a stronger collaborative culture.

This internship has been a transformative experience, bridging the gap between academic learning and professional practice. By reflecting on my accomplishments, challenges, and areas for improvement, I am better prepared to apply my knowledge and skills in future endeavors. The combination of hands-on project experience, exposure to industry-standard tools and methodologies, and mentorship from experienced professionals has provided me with a solid foundation for my career in software engineering.

4 SAMPLE WORK

This section provides examples of work-related artifacts produced during my internship, showcasing my contributions to the Mega Power Gym Management System and the PointSell POS System.

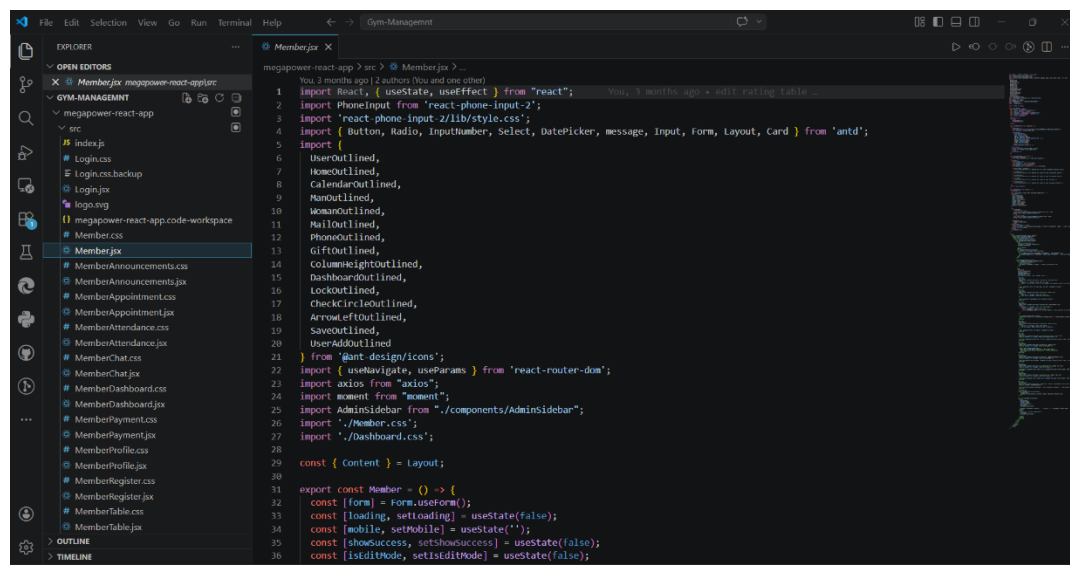
4.1 Mega Power Gym Management System – Frontend Screenshots

The Mega Power Gym Management System was developed as a complete full-stack web application using React.js (frontend) and Node.js/Express.js with MySQL (backend). The system replaces a fully manual gym management process involving paper registers and spreadsheets.

System Architecture Overview:

The application follows a three-tier architecture:

- Frontend: React.js with Material-UI components, Context API for state management
- Backend: Node.js with Express.js, following MVC pattern
- Database: MySQL with Prisma ORM for type-safe database operations



MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SQL File 30" students student_crud students students students students staffmember

1 • SELECT * FROM gym_management.staffmember;

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Staff_ID	FName	LName	DOB	Address	Gender	Contact_No	Email	Job_Role	UName
1	John	Trainer	1990-01-15 00:00:00	123 Main St, Colombo	Male	0777654321	john.trainer@megapowergym.com	Trainer	johntrai
2	Sarah	Instructor	1992-03-20 00:00:00	456 Park Ave, Colombo	Female	0777654322	sarah.instructor@megapowergym.com	Trainer	sarahins
3	Mike	Reception	1988-07-10 00:00:00	789 Lake Rd, Colombo	Male	0777654323	mike.reception@megapowergym.com	Receptionist	mikerece
4	Test Staff		1990-01-01 00:00:00	123 Test St	Male	0771234567	test.staff@example.com	Trainer	0771234
5	Dinithi Gamage		2007-01-29 18:30:00	554/15, Kandyiyadda Paluwa	Male	94778590364	sample123@email.com	Trainer	9477856

Table: staffmember

Columns:

- Staff_ID int(11) PK
- FName varchar(255)
- LName varchar(255)
- DOB datetime
- Address varchar(255)
- Gender varchar(50)
- Contact_No varchar(100)
- Email varchar(255)
- Job_Role varchar(100)
- UName varchar(255)
- Password varchar(255)
- Status varchar(50)
- Salary decimal(10,2)

Output

Action Output

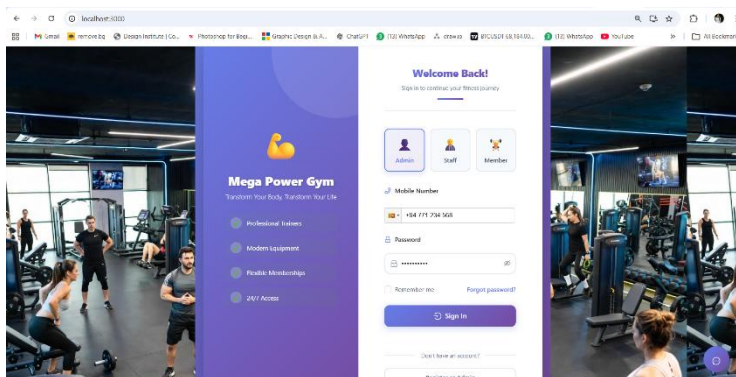
1 12:33:29 SELECT * FROM gym_management.staffmember LIMIT 0, 1000

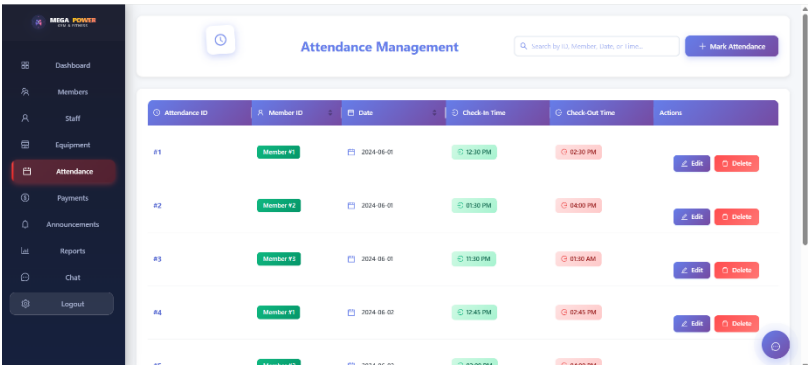
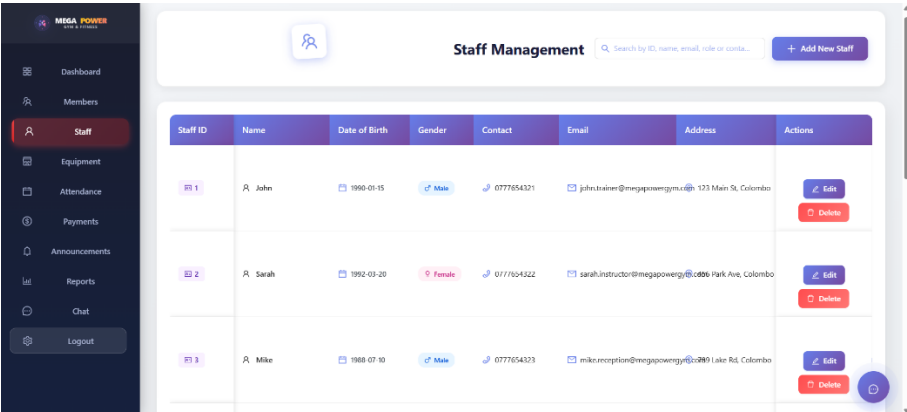
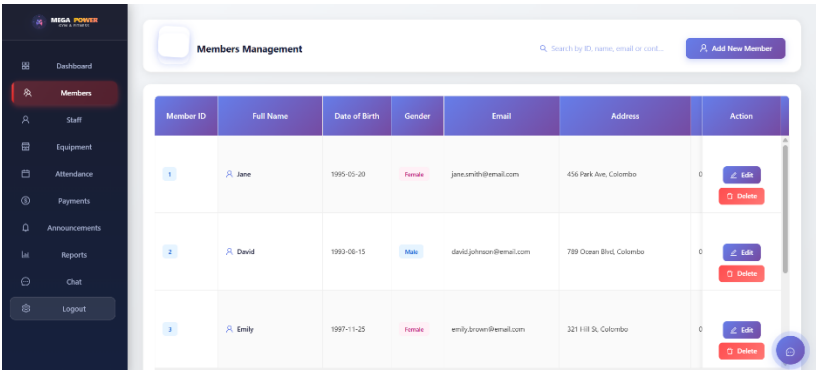
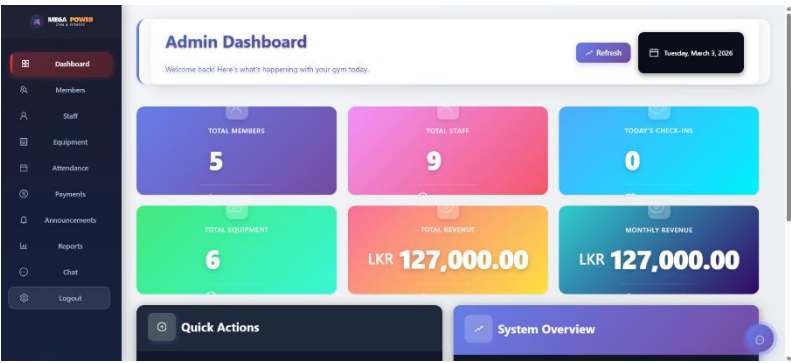
Message

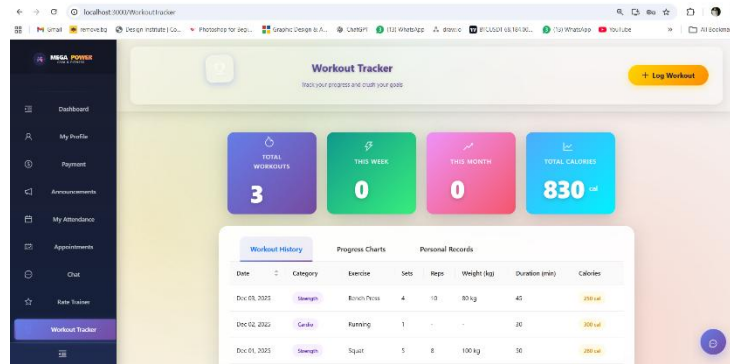
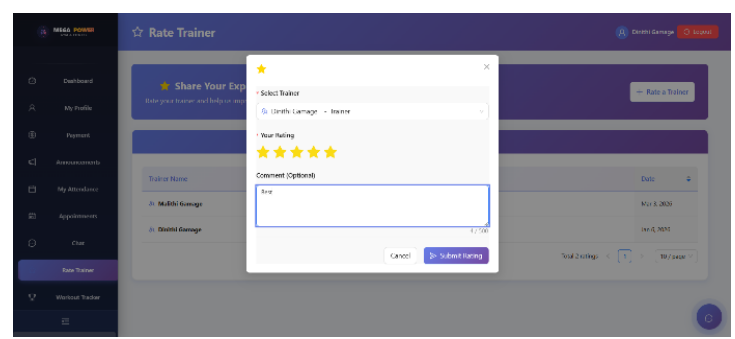
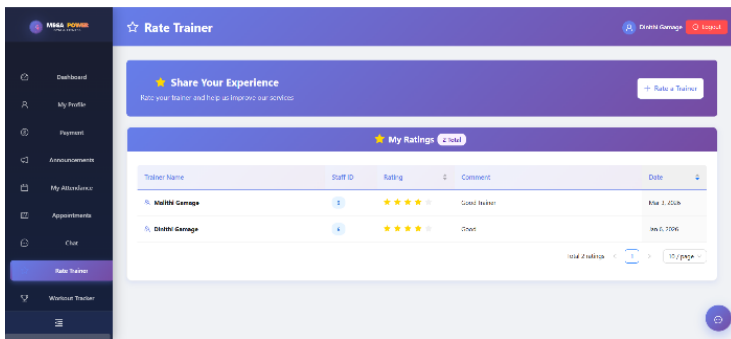
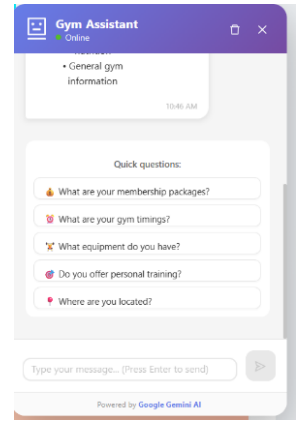
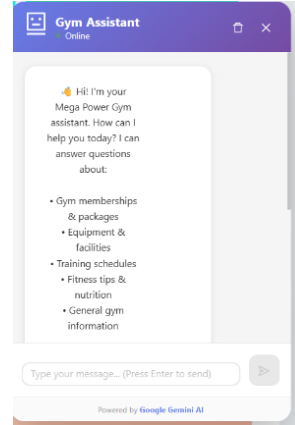
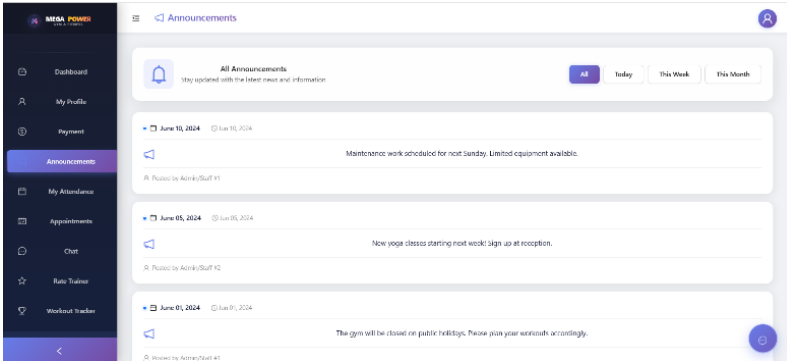
9 row(s) returned

Duration / Fetch

0.031 sec / 0.000 sec







4.2 PointSell POS System – API Architecture and Dashboard

The PointSell POS Web Application was developed as a team project. The backend was built using Node.js/Express.js with Sequelize ORM and MySQL 8.0, featuring 45+ RESTful API endpoints documented via Swagger/OpenAPI.



EXECUTIVE SUMMARY

1.1 Project Brief

This project aims to develop a modern, mobile-responsive Point of Sale (POS) system specifically designed for small and medium retailers in Sri Lanka. The system will address common challenges in existing POS solutions including reliability, offline capability, and local compliance requirements.

1.2 Key Features

- Offline-first Progressive Web Application (PWA)
- Real-time inventory management with batch tracking
- Multi-language support (Sinhala/Tamil/English)
- Sri Lanka VAT compliance and reporting
- Loyalty program and customer management
- Hardware-agnostic design (works on any device)

1.3 Project Duration

Total Duration: 24 Weeks (6 Months)

Start Date: 24.11.2025

End Date: 30.06.2025

1.4 Detailed System Architecture Diagram

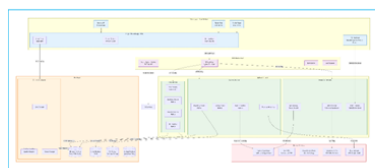


Figure 1: Detailed System Architecture



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2 PROJECT OVERVIEW

2.1 Project Background

Small and medium retailers in Sri Lanka face challenges with existing POS systems including frequent downtime, poor inventory management, and lack of local language support. This project addresses these gaps with a modern solution.

2.2 Problem Statement

1. Technical reliability issues during peak hours
2. Limited integration capabilities
3. Poor inventory tracking
4. Security and data protection risks
5. Hardware dependence and high maintenance
6. Complex user interfaces requiring extensive training
7. Scalability limitations

2.3 Solution Overview

A mobile-responsive web application built as a Progressive Web App (PWA) with offline-first architecture, designed specifically for Sri Lankan retail environment.

2.4 Project Constraints

- Budget: [Amount] LKR maximum
- Time: 6-month delivery timeline
- Resources: [Number] team members
- Technology: Must work on low-end devices common in Sri Lanka
- Compliance: Must meet Sri Lanka VAT and business regulations



2.5. User case Diagram

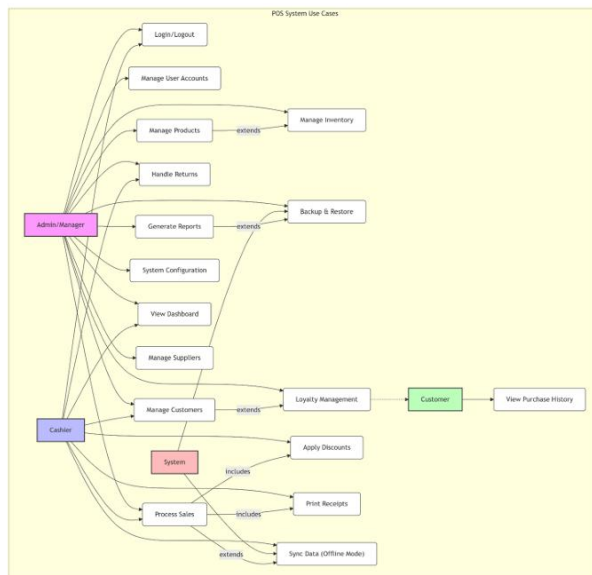


Figure 2: POS System UseCase Diagram

Inventory Inventory batch management with expiry tracking		
GET	/api/inventory/ingredients	Get all ingredients
POST	/api/inventory/ingredients	Create new ingredient
GET	/api/inventory/ingredients/{id}	Get ingredient by ID
PUT	/api/inventory/ingredients/{id}	Update ingredient
DELETE	/api/inventory/ingredients/{id}	Delete ingredient
PUT	/api/inventory/ingredients/{id}/stock	Update ingredient stock levels
POST	/api/inventory/wastage	Record product wastage
GET	/api/inventory/alerts	Get low stock alerts
PATCH	/api/inventory/alerts/{id}/acknowledge	Acknowledge stock alert
PATCH	/api/inventory/alerts/{id}/resolve	Resolve stock alert
POST	/api/inventory/reconciliation	Perform daily stock reconciliation
GET	/api/inventory/reports/daily-usage	Generate daily usage report
GET	/api/inventory/reports/wastage	Generate wastage report

Sales Sales processing with automatic 15% VAT calculation		
POST	/api/sales	Create new sale
GET	/api/sales	Get all sales
GET	/api/sales/report	Get sales report
GET	/api/sales/{id}	Get sales by ID
PATCH	/api/sales/{id}	Update sales status
POST	/api/sales/vat/calculate	Calculate VAT for items
GET	/api/sales/{id}/vat	Get VAT breakdown for sales
GET	/api/sales/reports/vat	Generate VAT compliance report
GET	/api/sales/reports/vat/{period}	Generate VAT report for specific period
GET	/api/sales/reports/vat/export/csv	Export VAT report to CSV
PATCH	/api/sales/{id}/status	Update order status
PATCH	/api/sales/{id}/cancel	Cancel order with reason
PATCH	/api/sales/{id}/complete	Complete order with optional reason

Cash Payments Cash payment processing and cash drawer management		
POST	/api/payments/cash/process	Process a cash payment transaction
POST	/api/payments/cash/validate	Validate cash denomination and amount
GET	/api/payments/cash/drawer	Get current cash drawer status
POST	/api/payments/cash/drawer/open	Open cash drawer for a sale
POST	/api/payments/cash/drawer/close	Close cash drawer and reconcile
POST	/api/payments/cash/refund	Process cash refund
POST	/api/payments/cash/drawer/reconcile	Reconcile cash drawer
GET	/api/payments/reconciliation/report	Get reconciliation report
GET	/api/payments/reconciliation/discrepancies	Get drawer discrepancies
GET	/api/payments/transaction	Get payment transaction history
GET	/api/payments/cash/drawer/summary	Get drawer summary statistics

Cash Payments Cash payment processing and cash drawer management		
POST	/api/payments/cash/process	Process a cash payment transaction
POST	/api/payments/cash/validate	Validate cash denomination and amount
GET	/api/payments/cash/drawer	Get current cash drawer status
POST	/api/payments/cash/drawer/open	Open cash drawer for a sale
POST	/api/payments/cash/drawer/close	Close cash drawer and reconcile
POST	/api/payments/cash/refund	Process cash refund
POST	/api/payments/cash/drawer/reconcile	Reconcile cash drawer
GET	/api/payments/reconciliation/report	Get reconciliation report
GET	/api/payments/reconciliation/discrepancies	Get drawer discrepancies
GET	/api/payments/transactions	Get payment transaction history
GET	/api/payments/cash/drawer/summary	Get drawer summary statistics

Tables		
GET	/api/tables/status/available	Get available tables
GET	/api/tables/status/occupied	Get occupied tables
POST	/api/tables	Create a new table
GET	/api/tables/{id}	Get table by ID
PUT	/api/tables/{id}	Update table
DELETE	/api/tables/{id}	Delete table
POST	/api/tables/{id}/assign	Assign an order to a table
POST	/api/tables/{id}/release	Release a table (mark as available)
GET	/api/tables/{id}/orders	Get all orders assigned to this table

Kitchen Kitchen order routing and management system		
POST	/api/kitchen/orders	Create a new kitchen order
GET	/api/kitchen/orders	Get active kitchen orders
GET	/api/kitchen/orders/{id}	Get kitchen order by ID
PUT	/api/kitchen/orders/{id}	Update kitchen order
PATCH	/api/kitchen/orders/{id}/status	Update kitchen order status
PATCH	/api/kitchen/orders/{id}/link-sale	Link a sale to a kitchen order after payment
PATCH	/api/kitchen/orders/{id}/items/{itemId}/status	Update item status in kitchen order
PATCH	/api/kitchen/orders/{id}/cancel	Cancel kitchen order with reason
PATCH	/api/kitchen/orders/{id}/delay	Notify delay for kitchen order
PATCH	/api/kitchen/orders/{id}/reset-delay	Reset delay notification for an order
GET	/api/kitchen/metrics	Get kitchen performance metrics
POST	/api/kitchen/stations	Create a new kitchen station
GET	/api/kitchen/stations	Get all kitchen stations

STATUS

LIVE SYSTEM

Search for orders, tables or products...

2/10

Admin

Admin

Dashboard

POS Terminal

Kitchen

Active Orders

Table Map

Inventory

Black

Menu

Admin

Login

Ayubowan,

It's Wednesday, March 4

CURRENT SESSION

Admin Mode

Quick Access Hub

POS

Quickly navigate to this section

Inventory

Quickly navigate to this section

Reports

Quickly navigate to this section

Menu

Quickly navigate to this section

Today's Performance

TOTAL REVENUE

Rs. 0.00

+12.5%

TOTAL ORDERS

0

+5.2%

ACTIVE CUSTOMERS

142

+2.4%

POS SYSTEM VALUE

Rs. 0.00

+8.1%

STATUS

LIVE SYSTEM

Search for orders, tables or products...

2/10

Admin

Admin

Dashboard

POS Terminal

Kitchen

Active Orders

Table Map

Inventory

Black

Menu

Admin

Login

ACTIVE STATION

Main Kitchen

Switch Station

Special Menu

Search products

kottu

Rs. 1200.00

+

pizza

Rs. 1400.00

+

Rice

Rs. 1200.00

+

Current Order

Subtotal

Rs. 1200.00

Tax

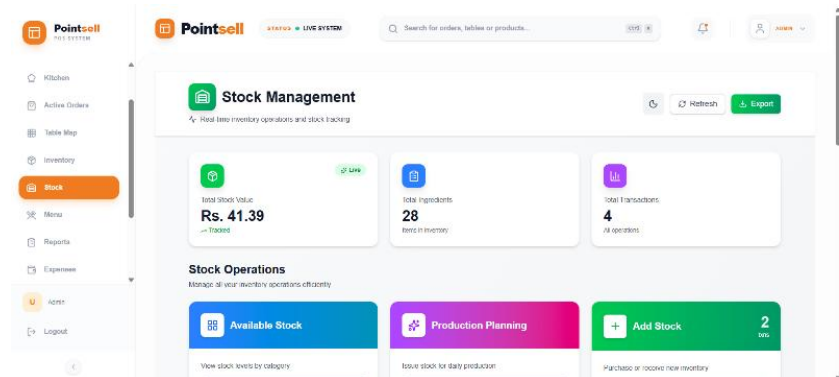
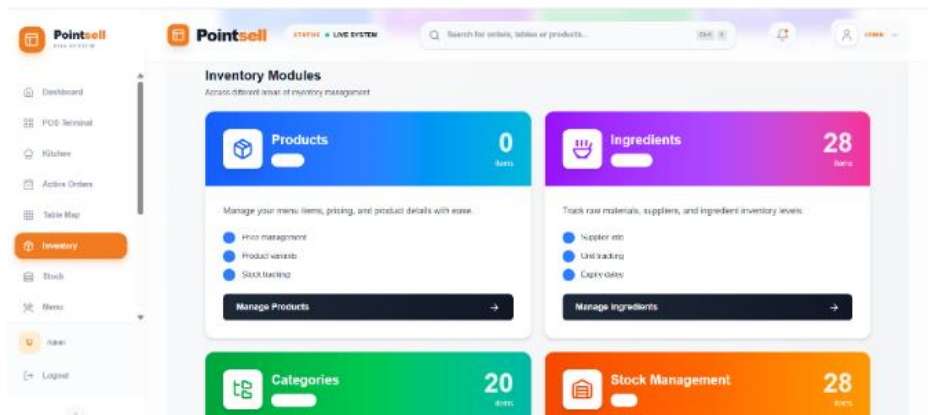
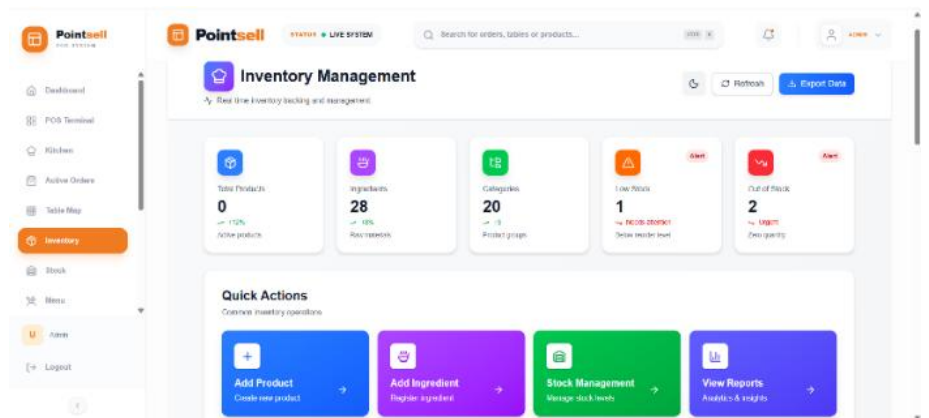
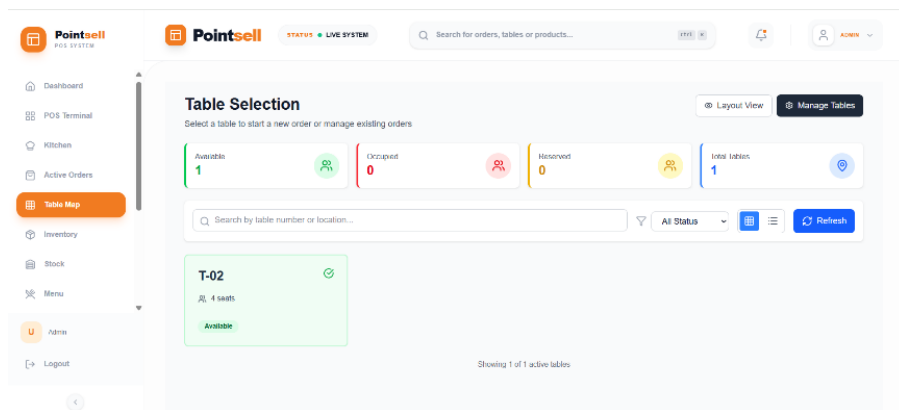
Rs. 180.00

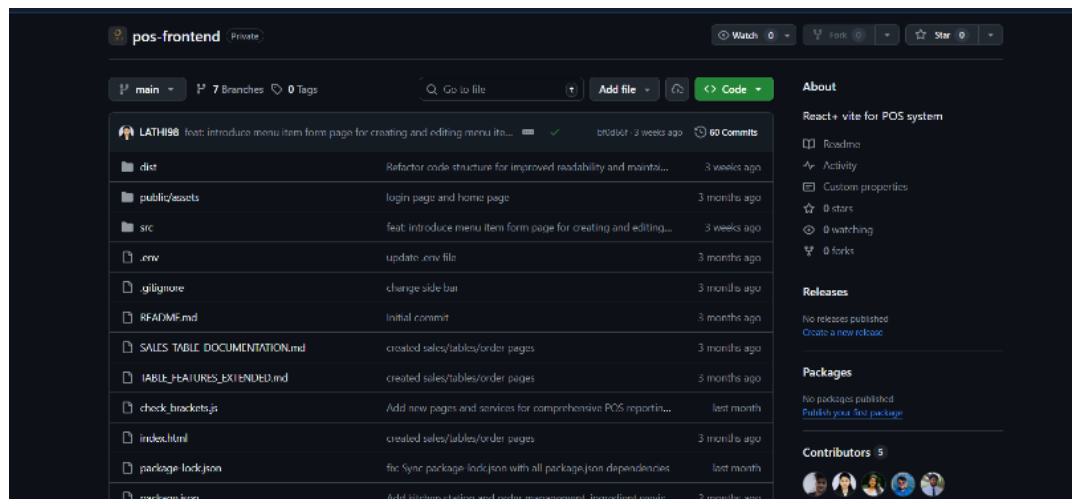
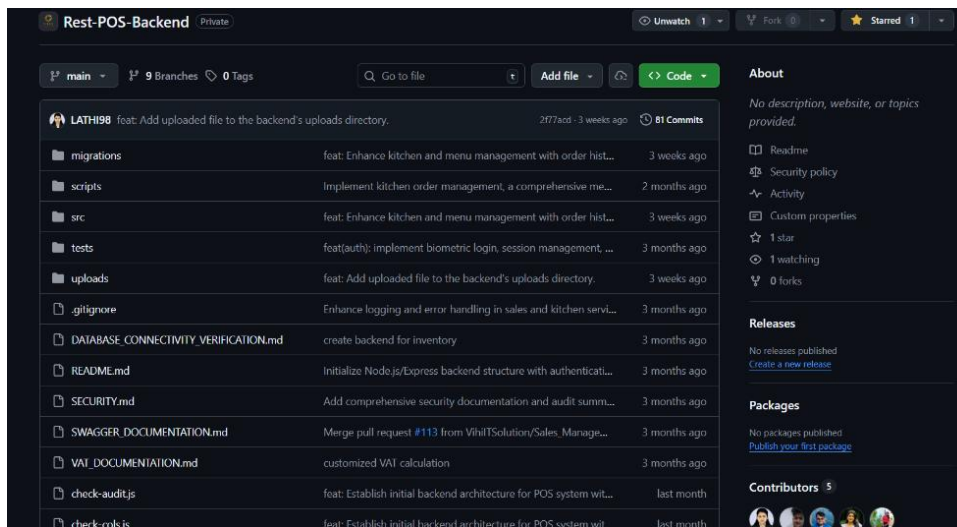
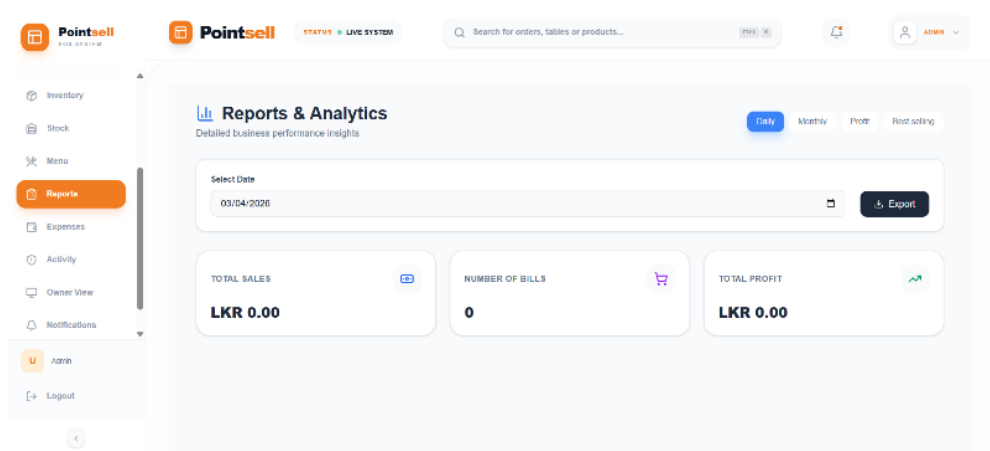
Total

Rs. 1380.00

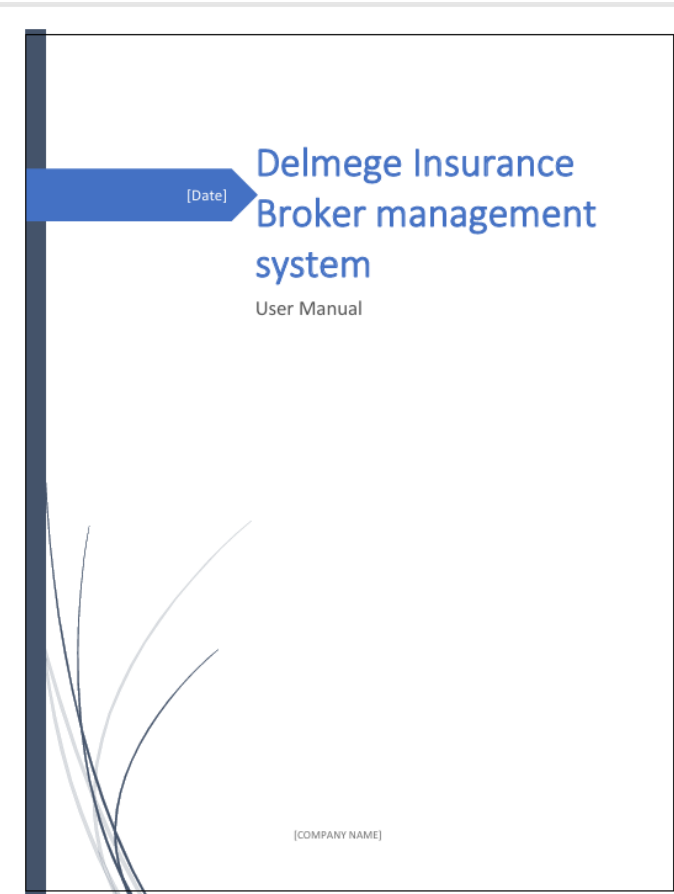
Proceed to Checkout

25 | Page





4.3 Delmege ERP System – User Manual Documentation



1. Introduction

About Delmege ERP

Delmege ERP is a comprehensive Enterprise Resource Planning solution designed specifically for broker companies and insurance management operations. This system integrates multiple business processes into a unified platform, enabling efficient management of companies, policies, sectors, client relationships, and insurance operations.

Purpose of This Manual

This user manual provides step-by-step instructions for navigating and utilizing the Delmege ERP system. It covers all major features, from basic navigation to advanced module management, ensuring users can maximize the system's capabilities effectively.

System Overview

The Delmege ERP system includes the following core modules:

- Dashboard - Real-time metrics and quick navigation
- RAQ Management - Request for quotation handling
- Sectors Management - Industry classification and tracking
- Policies Management - Insurance policy administration
- Client Companies Management - Customer relationship management
- Insurance Companies Management - Insurance provider management
- User Management - System access and permissions
- NatCat Exposure - Natural catastrophe risk assessment
- Settings - System configuration
- Jobs - Task and workflow management

System Access

To access the Delmege ERP system:

- Open your web browser (Chrome, Firefox, Safari, or Edge recommended)
- Navigate to the system URL
- Enter your username and password provided by your system administrator
- Click the login button to access the dashboard

User Interface Overview

The Delmege ERP interface consists of several key components designed for intuitive navigation and efficient workflow management.

- Top Navigation Bar

- Notification bell icon
- User profile dropdown (delmegeadmin)

The sidebar provides quick access to all system modules with visual icons and labels for easy identification.

1.1 Dashboard

The Dashboard is your central hub for monitoring key business metrics and accessing frequently used functions. It provides real-time insights into company performance, policy status, and system activities.

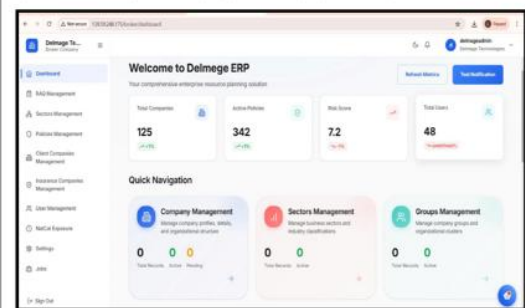


Figure 1-Delmege ERP Dashboard Overview

Key Metrics Cards

The top section of the dashboard displays four essential metrics that provide a quick snapshot of your business status:

Total Companies

- Current count: 125 companies
- Growth Indicator: +1% (green arrow showing positive trend)

- Shows the number of currently active insurance policies

Risk Score

- Current score: 7.2
- Trend indicator: -1% (red arrow showing slight decrease)
- Provides an aggregate risk assessment across all policies and companies

Total Users

- Current count: 48 users
- Status: undefined% (percentage change not calculated)
- Displays the total number of system users with access credentials

Quick Navigation Section

The Quick Navigation section provides direct access to frequently used management modules with real-time record counts:

Company Management

- Function: Manage company profiles, details, and organizational structure
- Total Records: 0
- Active: 0
- Pending: 0
- Click the arrow icon to navigate to the detailed management interface

Sectors Management

- Function: Manage business sectors and industry classifications
- Total Records: 0
- Active: 0

Groups Management

- Function: Manage company groups and organizational clusters
- Total Records:

Additional Dashboard Modules

Asset Codes

- Function: Manage asset classifications, codes, and inventory tracking
- Total Records: 0
- Active: 0
- Pending: 0

MIS Reports

- Function: Generate comprehensive management information system reports

5

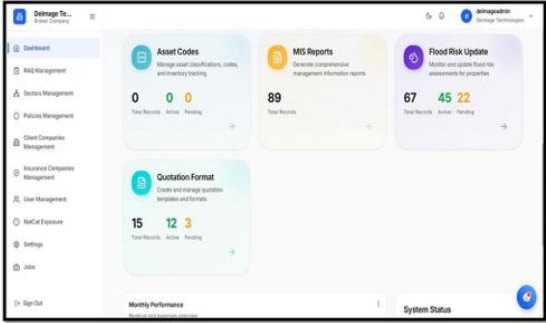
- Total Records: 89 reports available

Flood Risk Update

- Function: Monitor and update flood risk assessments for properties
- Total Records: 67
- Active: 45
- Pending: 22

Quotation Format

- Function: Create and manage quotation templates and formats
- Total Records: 15
- Active: 12
- Pending: 3



The screenshot shows a dashboard for 'Damagex Te...' with a sidebar menu on the left containing: Dashboard, MIS Management, Sectors Management, Policies Management, Client Companies Management, Insurance Companies Management, User Management, Net/GA Exposure, Settings, and Jobs. The main area displays four cards: 'Asset Codes' (0 Total, 0 Active, 0 Pending), 'MIS Reports' (89 Total), 'Flood Risk Update' (67 Total, 45 Active, 22 Pending), and 'Quotation Format' (15 Total, 12 Active, 3 Pending). At the bottom, there are sections for 'Monthly Performance' and 'System Status'.

Figure 2-Additional Dashboard Modules

System Status and Performance

Monthly Performance

The Monthly Performance widget provides a revenue and expenses overview for tracking financial metrics across time periods.

6